

Degeneracy and Orthogonalization of Eigenvectors

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1 Introduction

In quantum mechanics, the eigenvectors of an operator often form a natural basis for the Hilbert space. However, a complication arises when an operator possesses **degenerate eigenvalues**.

Degeneracy means that several independent eigenvectors correspond to the same eigenvalue. These eigenvectors span a subspace known as the **degenerate subspace**.

Although the eigenvectors are linearly independent, they are not necessarily orthogonal. Since orthogonal bases are extremely useful in quantum mechanics, we typically construct an orthogonal set of vectors within the degenerate subspace.

The standard method used to achieve this is the **Gram–Schmidt orthogonalization procedure**.

2 Degenerate Eigenvalues

Suppose an operator $\hat{A}$ satisfies the eigenvalue equation

$$\hat{A} |a_k\rangle = a_k |a_k\rangle. \quad (1)$$

If the same eigenvalue a corresponds to several eigenvectors,

$$|a_1\rangle, |a_2\rangle, \dots, |a_g\rangle,$$

then the eigenvalue is said to be **g -fold degenerate**.

This means

$$\hat{A} |a_j\rangle = a |a_j\rangle, \quad j = 1, 2, \dots, g. \quad (2)$$

The vectors $\{|a_j\rangle\}$ form a basis for the degenerate eigenspace, but they may not be orthogonal.

3 Need for Orthogonalization

In quantum mechanics, orthonormal bases are extremely convenient because they simplify calculations of probabilities, expectation values, and expansions of quantum states.

Therefore, when degenerate eigenvectors are not orthogonal, we construct a new set of vectors

$$\{|b_1\rangle, |b_2\rangle, \dots, |b_g\rangle\}$$

such that

$$\langle b_m | b_n \rangle = \delta_{mn}. \quad (3)$$

4 First Vector

The first orthogonal vector can be chosen freely from the degenerate eigenvectors:

$$|b_1\rangle = |a_1\rangle. \quad (4)$$

This vector is already an eigenvector of $\hat{A}$.

5 Second Vector

The second orthogonal vector is constructed as a linear combination of the original eigenvectors:

$$|b_2\rangle = |a_2\rangle + \lambda |b_1\rangle. \quad (5)$$

We impose the orthogonality condition

$$\langle b_1 | b_2 \rangle = 0. \quad (6)$$

Substituting the expression for $|b_2\rangle$:

$$\langle b_1 | a_2 \rangle + \lambda \langle b_1 | b_1 \rangle = 0. \quad (7)$$

Solving for λ gives

$$\lambda = -\frac{\langle b_1 | a_2 \rangle}{\langle b_1 | b_1 \rangle}. \quad (8)$$

Thus the second orthogonal vector becomes

$$|b_2\rangle = |a_2\rangle - |b_1\rangle \frac{\langle b_1 | a_2 \rangle}{\langle b_1 | b_1 \rangle}. \quad (9)$$

6 Third Vector

The third orthogonal vector is written as

$$|b_3\rangle = |a_3\rangle + \mu |b_1\rangle + \nu |b_2\rangle. \quad (10)$$

We impose two orthogonality conditions:

$$\langle b_1 | b_3 \rangle = 0, \quad (11)$$

$$\langle b_2 | b_3 \rangle = 0. \quad (12)$$

Solving these equations leads to

$$|b_3\rangle = |a_3\rangle - |b_1\rangle \frac{\langle b_1 | a_3 \rangle}{\langle b_1 | b_1 \rangle} - |b_2\rangle \frac{\langle b_2 | a_3 \rangle}{\langle b_2 | b_2 \rangle}. \quad (13)$$

7 General Gram–Schmidt Formula

In general, the k -th orthogonal vector is obtained from

$$|b_k\rangle = |a_k\rangle - \sum_{j=1}^{k-1} |b_j\rangle \frac{\langle b_j | a_k \rangle}{\langle b_j | b_j \rangle}. \quad (14)$$

This formula systematically removes components along previously constructed vectors.

8 Normalization

To obtain an orthonormal basis, each vector is normalized:

$$|\tilde{b}_k\rangle = \frac{|b_k\rangle}{\sqrt{\langle b_k|b_k\rangle}}. \quad (15)$$

The resulting set satisfies

$$\langle \tilde{b}_m | \tilde{b}_n \rangle = \delta_{mn}. \quad (16)$$

9 Preservation of Eigenvalues

An important property is that orthogonalization does not change the eigenvalue equation.

Since each $|b_k\rangle$ is constructed as a linear combination of vectors belonging to the same degenerate subspace,

$$\hat{A} |b_k\rangle = a |b_k\rangle. \quad (17)$$

Thus the orthogonalized vectors remain eigenvectors with the same eigenvalue.

10 Physical Interpretation

Degeneracy occurs frequently in quantum systems due to symmetry.

Examples include

- hydrogen atom energy levels
- angular momentum multiplets
- spin states

Within degenerate subspaces, the choice of basis is not unique. Orthogonalization allows us to construct a convenient basis that simplifies calculations.

11 Summary

- Degeneracy occurs when multiple eigenvectors correspond to the same eigenvalue.
- Degenerate eigenvectors may not be orthogonal.
- The Gram–Schmidt procedure constructs an orthogonal basis.
- Orthogonalized vectors remain eigenvectors of the operator.